

Loan Approval Prediction: Performance Comparison of Tree-based and Logistic Regression Model

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Abstract—This study explores the use of machine learning techniques on a Kaggle loan approval dataset to compare the performance of Logistic Regression, Random Forest, and XGBoost. After data cleaning, one-hot encoding, feature scaling, and a stratified train-test split, each model was optimized through cross-validated hyperparameter tuning. Evaluation using accuracy, F1-score, and ROC-AUC shows that tree-based models, particularly XGBoost, outperform the linear baseline. Feature importance analysis further identifies the key borrower and loan attributes that influence approval outcomes, demonstrating the value of machine learning for reliable credit assessment.

1. INTRODUCTION

1.1. Problem Definition

Loan approval is a critical task in financial institutions because it directly affects credit risk, operational efficiency, and customer outcomes. Traditional loan assessment methods like manual review can be slow, inconsistent, and limited in capturing complex relationships among applicant characteristics. As the volume of loan applications increases and financial decisions rely more heavily on data, machine learning provides a more accurate and scalable way to assess creditworthiness and support lending decisions.

The goal of this project is to build models that predict whether a loan application will be approved and to compare the performance of Logistic Regression with tree-based classification models. A successful solution would generate accurate, stable, and interpretable predictions that can be smoothly incorporated into real lending workflows. Such models benefit multiple stakeholders: lenders aiming to reduce risk, analysts seeking to understand key approval factors, and applicants who receive faster and more consistent evaluations.

1.2. Dataset Description

The dataset used in this study originates from Kaggle and represents a synthetic credit-risk dataset designed for loan approval analysis [1]. It contains 45,000 records and 14 variables spanning both continuous and categorical data types. With its comprehensive coverage of demographic attributes, financial capacity indicators, loan characteristics, and credit history information, the dataset is suitable for supervised learning tasks aimed at predicting loan approval outcomes.

The features can be grouped into four major categories: demographic characteristics (such as age, gender, and education

level), financial status (such as annual income and employment experience), loan-related attributes (such as ownership, loan amount, interest rate, loan intent, and loan-to-income ratio), and credit history variables (such as credit score, credit history length, and records of previous defaults). The target variable, loan status, is a binary indicator: 1 denotes an approved loan, and 0 denotes a rejected loan, making the task a standard binary classification problem.

The dataset still presents several limitations and potential sources of bias. As a synthetic dataset, some relationships among features may reflect artificially generated patterns rather than genuine borrower behavior. The dataset also contains data-quality issues, such as unrealistic age values (e.g., records exceeding 100 years old) and severe class imbalance in default-related variables. These issues require careful pre-processing and validation before modeling.

1.3. Contributions

In this study, supervised machine learning methods were applied to the loan approval dataset to build and compare predictive classification models. The workflow included removing unrealistic outliers, applying one-hot encoding and scaling to features, and using a stratified train-test split to maintain class balance. Logistic Regression, Random Forest, and XGBoost models were trained and optimized using GridSearchCV and RandomizedSearchCV to ensure fair performance comparison. Model evaluation was conducted using accuracy, F1-score, ROC-AUC, and confusion matrices comparisons.

2. LITERATURE REVIEW

Credit risk assessment and loan approval prediction are huge deals in the money world, where classifying good borrowers from bad ones is essential to maintain institutional stability and mitigate risk. Machine learning models are increasingly replacing traditional statistical and rule-based systems in this sector.

2.0.1. Benchmarking Ensemble Methods in Default Prediction: Our reading shows that the performance of tree-based ensemble models is more superior to conventional linear approaches, such as Logistic Regression. Wu (2022) conducted a direct comparison of the prominent ensemble techniques, Random Forest and extreme Gradient Boosting (XGBoost), in

predicting loan defaults [4]. The study concluded that both RF and XGBoost achieve high and robust predictive accuracy, typically reaching above 90% in real-world lending scenarios, indicating these models perform well in credit risk analysis.

2.0.2. Automation, Model Comparison, and Applicability: Besides prediction, ML models are utilized to automate and enhance the loan approval pipeline. Research by Sirishma et al. (2023), which compared nine different classifiers including RF and Decision Trees, confirmed that automated systems significantly accelerate credit processing, increase decision accuracy, and critically, reduce human bias by systematically evaluating large datasets [3]. Because of this kind of broad comparison, we are choosing to stick with the high-performing ensemble models instead of those simpler options.

2.0.3. Interpretability, Evaluation Metrics, and The Research Gap: In the highly regulated financial sector, model decisions must be transparent and justifiable. Qi (2025) focused specifically on the application of XGBoost and emphasized that while the model yields high accuracy, its true value lies in Feature Importance Analysis [2]. This interpretability is vital for financial institutions to comply with regulations and provide clear explanations for loan outcomes. Furthermore, given the severe class imbalance inherent in default datasets (non-defaulters vastly outnumber defaulters), evaluation must prioritize multi-dimensional metrics such as the Area Under the ROC Curve, Precision, Recall, and F1-Score, rather than relying solely on accuracy. Our study addresses these identified needs by rigorously benchmarking these ensemble models on a publicly available dataset, providing both a comprehensive multi-metric performance analysis and a detailed feature importance breakdown to directly inform practical risk management strategies.

3. METHODOLOGY

3.1. Model Overview

This study applies three machine learning models with increasing levels of complexity to predict loan approval status.

First, Logistic Regression is chosen as the baseline model. It is a standard statistical method used for binary classification tasks. This model estimates the probability of an outcome based on input features. Although it assumes linearity which might not fit complex financial data perfectly, it has significant advantages: It is fast to train and easy to interpret. It allows us to clearly understand how factors like income or credit score influence the loan decision.

The second model is Random Forest, representing a moderate level of complexity. This is an ensemble method that builds a large number of decision trees during training. Instead of relying on a single tree, it combines the predictions from many trees to make a final decision. This approach helps to reduce overfitting, where the model memorizes the training data. Compared to Logistic Regression, Random Forest is much better at capturing complicated patterns and nonlinear relationships in the dataset.

The third model is XGBoost, which is the most advanced model in this study. XGBoost uses gradient boosting where

trees are built in a sequence, meaning each new tree tries to correct the mistakes made by the previous ones. It includes features to prevent the model from becoming too complex and is highly optimized for performance. This model serves as the upper benchmark in our comparison, aiming for the highest accuracy.

3.1.1. Code Architecture: Our code follows a structured workflow consisting of four main stages:

- (1) Data loading and Exploratory Data Analysis
- (2) Pre-processing and label encoding
- (3) Model training and tuning
- (4) Evaluation and comparison of models' performance

We first load the Loan Approval Classification Dataset [1] from the Kaggle Repository and examine its characteristics through exploratory data analysis. After pre-processing the data, we train the three selected models in a controlled and consistent manner so their performance can be fairly compared. The final outputs include predictions, evaluation metrics, and visual explanations.

3.2. Exploratory Data Analysis (EDA)

	person_age	person_income	person_emp_exp	loan_amnt	loan_int_rate	loan_percent_income	cb_person_cred_hist_length	credit_score
count	45000.000000	4.500000e+04	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000	45000.000000
mean	27.764176	8.031905e+04	5.410333	9563.157556	11.006606	0.139725	5.867489	632.608756
std	6.045108	8.042250e+03	6.063532	6314.886691	2.978808	0.087212	3.879702	50.435865
min	20.000000	8.000000e+03	0.000000	500.000000	5.420000	0.000000	2.000000	390.000000
25%	24.000000	4.720400e+04	1.000000	5000.000000	8.580000	0.070000	3.000000	601.000000
50%	26.000000	6.704800e+04	4.000000	8000.000000	11.010000	0.120000	4.000000	640.000000
75%	30.000000	9.578925e+04	8.000000	12237.250000	12.990000	0.190000	8.000000	670.000000
max	144.000000	7.200766e+06	125.000000	35000.000000	20.000000	0.660000	30.000000	850.000000

Fig. 1. Descriptive statistics of numerical features.

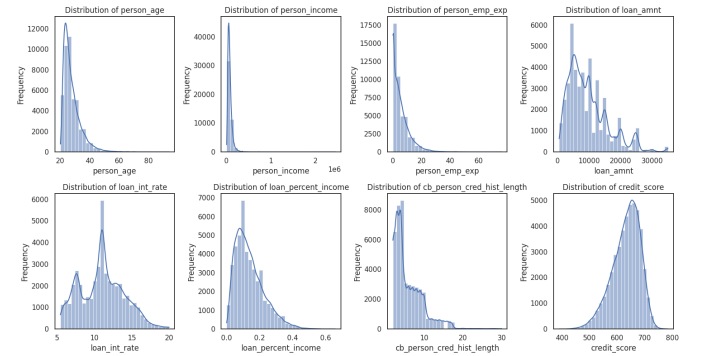


Fig. 2. Histograms and kernel density estimates for all numerical variables.

We conducted EDA on the 45,000 records. Descriptive statistics (Figure 1) were first applied to all numerical features to understand the basic distribution of variables and identify potential anomalies.

Results indicate significant dispersion in variables such as age, income, and loan amount. Age exhibits some outliers with a maximum value of 144 years, and work experience contains an extreme value of 125 years. These anomalies suggest unreasonable inputs in the dataset, providing clues for data cleansing. There are no missing records in this dataset, so no additional imputation steps are required.

Furthermore, histograms and kernel density estimates were used to display the distribution of numerical variables as in Figure 2. The image and statistical results reveal that most variables in this dataset exhibit highly skewed distributions. Samples with smaller loan amounts are heavily clustered. While high-value loan samples form a long tail, which is a classic distribution characteristic of consumer loan data. The correspondence between age and work experience is also reflected in the distribution plots, showing a reasonable growth trend. The multi-peak distribution of loan interest rates further suggests this variable may be influenced by categorical factors such as loan purpose or risk rating.

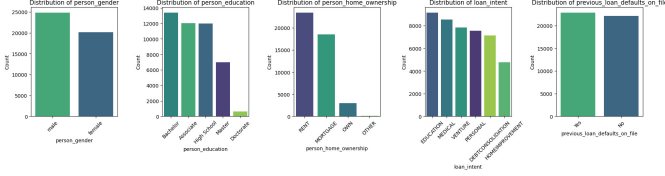


Fig. 3. Distribution of Categorical Features

In addition to numerical features, we analyzed the distribution of categorical variables to understand the demographic and loan-specific characteristics of the applicant pool. As shown in Figure 3, the dataset exhibits a balanced gender distribution. Education levels are diverse, with a significant proportion of applicants holding Bachelor's or Associate degrees. In terms of home ownership, "Rent" and "Mortgage" are the dominant categories. The loan intent distribution reveals that "Education" and "Medical" expenses are among the most common reasons for borrowing. Notably, the history of previous loan defaults shows a healthy class balance, providing valuable historical risk information for the models.

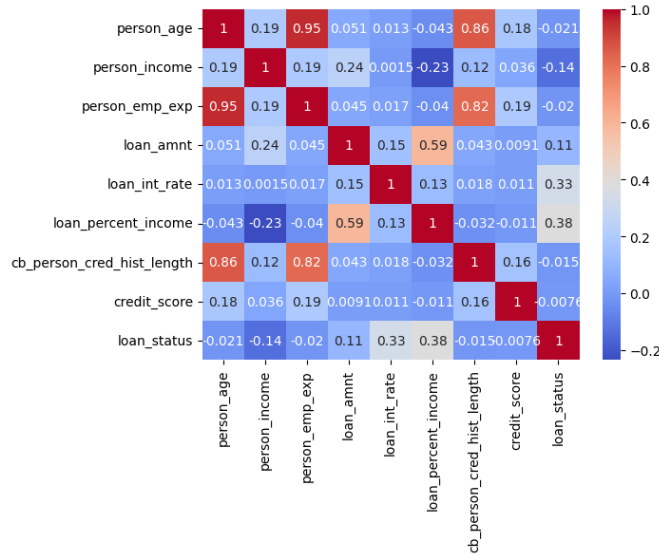


Fig. 4. Correlation heatmap of numerical features

In the visualization analysis section, the correlation heatmap (Figure 4) reveals structural relationships among certain vari-

ables. For instance, moderate correlation exists between loan amount and income ratio, while age and work experience show a clear positive correlation. These patterns provide insights for assessing variable redundancy and model explainability.

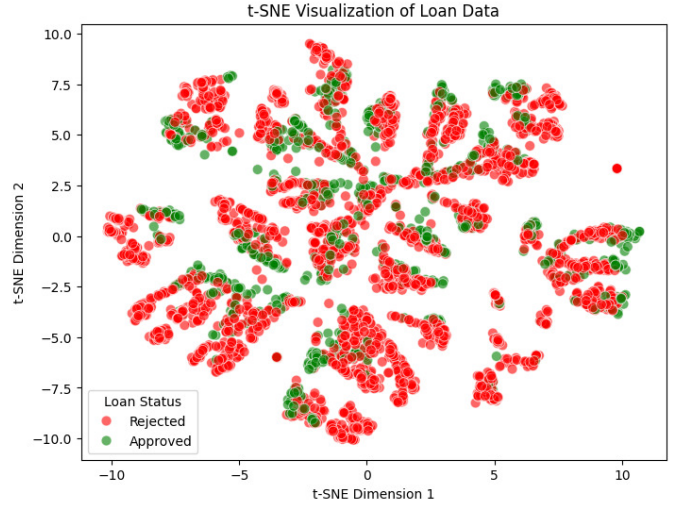


Fig. 5. t-SNE Visualization of Load Data

To further investigate the separability of the classes within the high-dimensional feature space, we employed t-Distributed Stochastic Neighbor Embedding (t-SNE) for dimensionality reduction. A stratified random sample of 5,000 instances was drawn from the standardized test set to generate a two-dimensional projection. As illustrated in Figure 5, the visualization reveals a complex, non-linear distribution where the "Approved" (Green) and "Rejected" (Red) classes are intertwined rather than linearly separable. This structural complexity suggests that linear classifiers may struggle to define an effective decision boundary, whereas non-linear ensemble methods (Random Forest and XGBoost) are theoretically better suited to capture these intricate patterns.

3.3. Data Cleaning and Pre-processing

3.3.1. Outlier Treatment: We first addressed data quality by handling extreme outliers in numerical features. Notice that for `person_age` there are some values that are larger than 100, which is unrealistic as human lives are not that long. Since the number of error entries was trivial, we deleted the rows with outliers, and there were 44993 rows left after processing. We also discovered that outliers in `person_emp_exp` were removed after above processing.

3.3.2. Feature Encoding and Transformation: As machine learning models require numerical input, all nominal categorical features, such as `person_gender`, `person_education`, need to be converted into a binary format using One-Hot Encoding. We employed the `drop_first = True` strategy during encoding to reduce multicollinearity among the dummy variables, enhancing the stability of the subsequent model training.

3.3.3. Data Splitting and Stratification: The dataset was divided into a 80% training set (X_{train}, y_{train}) for model development, and a 20% test set (X_{test}, y_{test}) for final performance assessment. `random_state = 42` was set for splitting and tuning to ensure the reproducibility of results. During the split, we applied stratified sampling (`stratify = y`). By doing this, the ratio of the target classes is identical in both the training and test sets, which guarantees the reliability and fairness of the evaluation metrics across the model comparison.

3.3.4. Feature Scaling: To prevent numerical features with large ranges from dominating the objective function, all numerical features were standardized. The `StandardScaler` was utilized to transform the features so that they followed a standard normal distribution with a mean (μ) of zero and a standard deviation (σ) of one. We only fit the scaler on the training set (X_{train}) to derive the statistical parameters (μ and σ), and then apply these derived parameters to transform both the training and test sets. This measure prevents data leakage, ensuring the model generalizes effectively to unseen data.

3.4. Modeling Approach

3.4.1. Model Training: All three models were trained using the same preprocessed training dataset to ensure a fair comparison. As shown in Figure 6, the dataset is imbalanced with more rejected (Class 0) loans than approved (Class 1), thus we applied specific techniques during training to handle this issue. For Logistic Regression and Random Forest, we used the "balanced" class weight setting. For XGBoost, we calculated the ratio between negative and positive samples to adjust the scale weight.

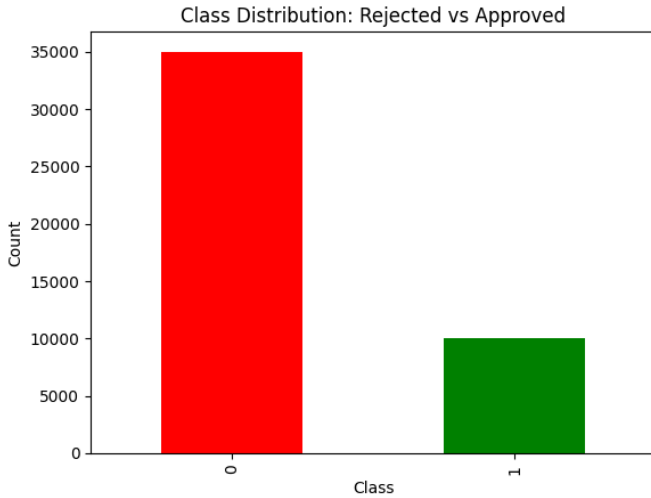


Fig. 6. Imbalanced Distribution of Target loan_status

3.4.2. Hyperparameter Tuning: We performed hyperparameter tuning to optimize all three models and to ensure a robust comparison.

For the Logistic Regression model, the hyperparameters considered include `C` and `solver`. The parameter `C` controls the inverse of regularization strength, where smaller values

specify stronger regularization to prevent overfitting. The solver determines the algorithm used for optimization.

For the Random Forest model, the hyperparameters considered include `n_estimators`, `max_depth`, `min_samples_split`, and `max_feature`. These parameters mainly control the number of trees in the ensemble and the complexity of each individual tree, which together influence the model's ability to capture non-linear patterns without memorizing the training data.

For the XGBoost model, the tuned hyperparameters include `learning_rate`, `n_estimators`, `max_depth`, `subsample`, and `gamma`. These parameters jointly control the step size of the boosting process, the complexity of the trees, and the sampling behaviour during training. This balance is crucial for minimizing the loss function while maintaining generalization capability.

To determine the optimal configuration, we adopted two cross-validation strategies. For Logistic Regression, we used Grid Search, which systematically evaluates every combination of parameters. However, for Random Forest and XGBoost, we used Randomized Search. This method randomly samples a fixed number of parameter settings, allowing for an efficient exploration of the large parameter space without excessive computational cost. In all cases, 5-fold cross-validation was used, and the F1-score was selected as the optimization metric to account for class imbalance.

Based on the cross-validation results, the best hyperparameters for Logistic Regression were:

- `C = 10`,
- `solver = "liblinear"`.

For Random Forest, the best hyperparameters selected were:

- `n_estimators = 300`,
- `max_depth = 20`,
- `min_samples_leaf = 2`
- `max_feature = log2`.

For XGBoost, the best hyperparameters selected by cross-validation were:

- `max_depth = 10`,
- `n_estimators = 500`,
- `learning_rate = 0.05`,
- `subsample = 0.8`
- `min_child_weight = 2`
- `gamma = 0.2`.

4. RESULTS AND DISCUSSION

4.1. Evaluation and Metrics

Given the significant target class imbalance in the loan dataset (approximately 78% rejected vs. 22% approved), relying on training and test accuracy is misleading. Therefore, in this study, classification report and confusion matrix are prioritized to ensure a fair and meaningful evaluation among three models. Specifically, we focus on the following metrics:

Recall for Class 1: the critical metric for capturing market share. It measures the percentage of actual qualified applicants

who are correctly identified. Low recall implies high opportunity costs, as the bank is rejecting customers who would have been profitable.

Precision for Class 1: measures the reliability of an "Approved" prediction. Low precision implies high financial risk, as it indicates the model is incorrectly approving unqualified applicants who are likely to default.

F1-Score: the harmonic mean of Precision and Recall, providing a balanced view of the model's performance in identifying qualified applicants while managing risk.

Area Under the ROC Curve (AUC): measures the model's overall ability to distinguish between qualified and unqualified applicants across all possible decision thresholds. Unlike F1-score, a high AUC indicates that the model is robust and effective at ranking applicants by risk level, providing flexibility for future adjustments in credit policy.

4.2. Comparative Analysis of Model Performance

The performance of the three models: Logistic Regression, Random Forest, and XGBoost, varied drastically, illustrating the impact of algorithm selection on imbalanced data.

4.2.1. Logistic Regression: The Logistic Regression model served as a baseline for this study, yielding an accuracy of 0.89 and an AUC score of 0.9522 (Figure 7). While these results are acceptable, the model showed a distinct performance gap compared to the tree-based ensemble methods. Specifically, its predictive power for approved loans (Class 1) was limited, with F1-score of only 0.76. This score was constrained by a Recall of 0.75 and a Precision of 0.77, indicating that the linear model missed approximately 25% of qualified applicants while achieving only moderate success in filtering risky borrowers.

```

--- Logistic Regression Classification Report ---
      precision    recall  f1-score   support

     0       0.93      0.94      0.93       6999
     1       0.77      0.75      0.76       2000

 accuracy          0.89       0.89       0.89       8999
 macro avg          0.85      0.84      0.85       8999
 weighted avg          0.89      0.89      0.89       8999

AUC Score: 0.9522

```

Fig. 7. Logistic Regression Classification Report

4.2.2. Random Forest: This model demonstrated robust performance, rivaling XGBoost with an identical overall accuracy of 0.93 and a competitive AUC score of 0.9743 (Figure 8). While its Recall for Class 1 (0.83) was lower than that of XGBoost, it achieved a marginally higher Precision (0.85). This indicates that while Random Forest may miss slightly more positive cases than XGBoost, it is more reliable when it does flag a positive case. It remains a very strong candidate, offering a balanced trade-off between precision and recall.

```

--- Random Forest Classification Report ---
      precision    recall  f1-score   support

     0       0.95      0.96      0.95       6999
     1       0.85      0.83      0.84       2000

 accuracy          0.93       0.93       0.93       8999
 macro avg          0.90      0.89      0.90       8999
 weighted avg          0.93      0.93      0.93       8999

AUC Score: 0.9743

```

Fig. 8. Random Forest Classification Report

4.2.3. XGBoost: The XGBoost model stands out as the top performer among the evaluated classifiers. It achieved the highest AUC score of 0.9786, indicating superior discriminative power in distinguishing between the two classes. Notably, XGBoost excelled in identifying the minority class (Class 1), achieving a recall of 0.87 and an F1-score of 0.85 in Figure 9. This high recall suggests the model is particularly effective at minimizing false negatives, making it the optimal choice if the business objective prioritizes capturing as many positive instances as possible.

```

--- XGBoost Classification Report ---
      precision    recall  f1-score   support

     0       0.96      0.95      0.96       6999
     1       0.83      0.87      0.85       2000

 accuracy          0.93       0.93       0.93       8999
 macro avg          0.90      0.91      0.90       8999
 weighted avg          0.93      0.93      0.93       8999

AUC Score: 0.9786

```

Fig. 9. XGBoost Classification Report

4.2.4. Confusion Matrix: The confusion matrices in Figure 10 provide a granular comparison of how each model handles correct predictions versus errors.

- **Logistic Regression:** This model exhibits the highest error rates across the board. Notably, it misclassified 507 "Approved" cases as "Rejected", indicating a significant loss of potential positive outcomes compared to the other models.
- **Random Forest:** This model proved to be the most conservative. It achieved the highest number of True Negatives (6700) and the lowest number of False Positives (299), meaning it is the least likely to incorrectly approve a rejected candidate. However, it missed more approved cases (343 False Negatives) than XGBoost.
- **XGBoost:** XGBoost demonstrated superior performance in identifying "Approved". It correctly identified the highest number of approved cases (1742 True Positives) and achieved the lowest number of False Negatives (258). Although it had slightly more False Positives (355) than Random Forest, the significant reduction in missed ap-

provals makes it the most effective model for maximizing the detection of positive instances.

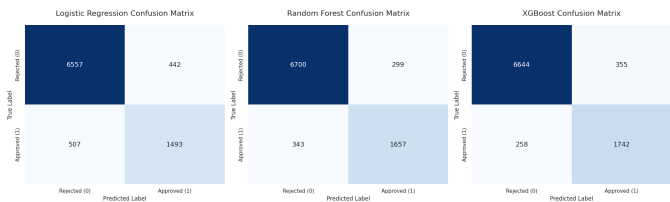


Fig. 10. Confusion Matrices for Logistic Regression, Random Forest, and XGBoost.

4.3. Model Selection and Justification

After a comprehensive evaluation of three classifiers, XGBoost has been selected as the final model. It demonstrated superior predictive performance, achieving the highest AUC score (0.9786) and Class 1 Recall (0.87). While Random Forest showed comparable overall accuracy, XGBoost proved significantly more effective at correctly identifying the minority class, which is a critical requirement for this use case. Conversely, the lower performance observed in Logistic Regression suggests that the dataset contains complex, non-linear patterns that are best captured by the advanced gradient boosting capabilities of XGBoost.

5. INTERPRETATION AND INSIGHTS

5.1. Conclusions

This study conclusively identifies XGBoost as the superior modeling technique for this classification task, achieving a peak AUC score of 0.9786 and an overall accuracy of 93%. The comparative analysis revealed that tree-based ensemble methods (XGBoost and Random Forest) significantly outperformed the Logistic Regression baseline, indicating the presence of complex, non-linear patterns within the data that linear models failed to capture. While Random Forest demonstrated strong stability, XGBoost provided the critical advantage of higher sensitivity, achieving a Recall of 0.87 for the minority class compared to 0.83 for Random Forest. This 4% gain in identifying positive instances implies that XGBoost is better suited for minimizing false negatives, which is a crucial requirement for risk-sensitive applications. Consequently, XGBoost is selected as the final model, offering the best strategic balance between high detection rates and operational precision.

5.2. Limitations

Despite the strong predictive performance of the selected model, the following limitations should be acknowledged:

(1) Class Imbalance

The dataset exhibits a moderate class imbalance (approx. 7:2 ratio between Reject and Approve). While XGBoost handled this well, the model is still biased towards the majority class. There remains a risk that performance could degrade if the proportion of the minority class drops further in real-world production data.

(2) Model Interpretability

The choice of XGBoost prioritizes predictive power over explainability. Unlike Logistic Regression, which offers transparent feature coefficients, XGBoost is a "black-box" model. This makes it more challenging to explain the specific rationale behind individual predictions to non-technical stakeholders.

(3) False Negatives

Although XGBoost achieved a high recall of 0.87, it still misses approximately 13% of the positive instances. Depending on the cost of a missed detection, such as in fraud, this error rate may require further calibration or threshold adjustment.

APPENDIX A

This appendix describes the specific roles and contributions of each group member to this project.

- Jialin Cai: responsible for model construction and technical validation. Selected the three machine learning algorithms based on loan dataset and wrote the Python code. Performed debugging and validated the output results for correctness. Conducted hyperparameter tuning to optimize model performance. In the report, wrote the Methodology section regarding model selection and tuning strategies.
- Rui Deng: focused on theoretical background and project management. Conducted the literature review to find relevant studies on credit risk assessment. Set up communication channels and managed document sharing for collaboration. For the report, contributed to the Literature Review and Conclusion sections.
- Haien Lin: in charge of result analysis and quality control. Interpreted performance metrics and analyzed confusion matrices to compare models. Wrote the Results and Discussion section. Acted as the final editor, proofreading the entire report and revising the text for clarity.
- Wenqing Zhao: responsible for data management. Searched for and selected the appropriate loan approval dataset. Performed initial data analysis and EDA, and preprocessed including data cleaning and feature encoding. Contributed to the report by writing the sections on Introduction, Data Pre-processing and EDA.

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